

MySQL Database Queries & Setup Guide

Cloud Computing Student Deployment Lab | Flask + MySQL Project

1. Overview & Setup Instructions

This document contains the exact MySQL database schema, table creation queries, seed data, and verification statements required for students deploying the Flask + MySQL web application.

2. Complete MySQL Database Creation Script

Copy and execute the following SQL statements in MySQL Command Line Interface (CLI), MySQL Workbench, or phpMyAdmin:

```
-- =====
-- Flask + MySQL Student Deployment Test - Database Setup
-- =====

-- 1. Create Database
CREATE DATABASE IF NOT EXISTS `cloud_test_db`
DEFAULT CHARACTER SET utf8mb4
COLLATE utf8mb4_unicode_ci;

USE `cloud_test_db`;

-- 2. Create Users Table
CREATE TABLE IF NOT EXISTS `users` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `username` VARCHAR(50) NOT NULL UNIQUE,
  `email` VARCHAR(100) NOT NULL UNIQUE,
  `password_hash` VARCHAR(255) NOT NULL,
  `role` ENUM('user', 'admin') DEFAULT 'user',
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 3. Seed Initial Administrator User
-- Default Credentials: Username: admin | Password: admin123
INSERT INTO `users` (`username`, `email`, `password_hash`, `role`)
VALUES (
  'admin',
  'admin@cloudtest.com',
  'script:32768:8:1$7nXZwQ3p6Yp7$24c883ed6df65ecf50a8b9eeb2db8fa0b555d4ee7e3fa4923e5904d9c791dd15e34743',
  'admin'
)
ON DUPLICATE KEY UPDATE `username`=`username`;
```

3. Table Schema Structure (`users`)

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique record identifier

username	VARCHAR(50)	NOT NULL, UNIQUE	Student / User login name
email	VARCHAR(100)	NOT NULL, UNIQUE	User email address
password_hash	VARCHAR(255)	NOT NULL	Werkzeug secure password hash
role	ENUM('user','admin')	DEFAULT 'user'	Role authorization level
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Registration timestamp

4. Student Verification SQL Queries

Students can run these queries to test and verify database operations during lab evaluation:

```
-- Query 1: View all registered users
SELECT id, username, email, role, created_at FROM users;

-- Query 2: Count total registered users
SELECT COUNT(*) AS total_students FROM users WHERE role = 'user';

-- Query 3: Search user by username or email
SELECT * FROM users WHERE username = 'admin' OR email = 'admin@cloudtest.com';

-- Query 4: Reset password manually (if needed)
UPDATE users SET password_hash = '<new_hash>' WHERE username = 'student1';

-- Query 5: Delete test user account
DELETE FROM users WHERE username = 'test_user';
```